higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PLUMBING
NQF LEVEL 4

(12030544)

22 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 5 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only BLUE or BLACK ink.
6. Write neatly and legibly.

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QUESTION 1: BASIC BUILDING WORK AND MANHOLES

1.1 With the aid of a diagram, draw a neat sketch showing THREE layers of brick units in stretcher bond. (3)

1.2 Give a description of a manhole and state what the suggested depth for it should be. (2)

1.3 When the excavation for a new manhole has been completed the ground needs to be compacted before the concrete floor can be cast.

Give a description of the minimum thickness and strength for the concrete blinding on the floor which will be placed on the compacted ground. (4)

1.4 In plumbing terms, what does a *soffit* refer to? (1)

[10]

QUESTION 2: BELOW GROUND DRAINAGE SYSTEM

2.1 What does the term *rodding eye* refer to? (2)

2.2 Name TWO precautionary measures that should be in place to ensure the safety of people moving around or near an excavation at night. (2)

2.3 Who determines the site datum on a building site? (1)

2.4 What are the most effective gradients for drain pipes with the following diameters:

2.4.1 100 mm Ø

2.4.2 150 mm Ø

2.4.3 225 mm Ø

(3 × 1) (3)

2.5 Name FIVE levelling instruments that can be used during the excavation of trenches, with the intent to obtain the required invert levels? (5)

2.6 List TWO facts related to the term "*bedding.*" (2)

2.7 What is a difference between an *air-test* and a *water test*? (2)

2.8 Name THREE manuals that should be consulted when laying plumbing pipes. (3)

[20]

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QUESTION 3: PLUMBING MAINTENANCE AND REPAIRS

3.1 What is the purpose of a tap flange? (1)

3.2 List at least NINE steps that need to be followed during the repairing process of a leaking copper pipe. (9)

3.3 What is the function of a thermostat, situated in a geyser? (1)

3.4 Name the FOUR procedures (in sequential order) that should be followed when replacing a pressure-relief valve. (4)

3.5 List FIVE methods that can be used to remove obstructions or blockages in a drainage system. (5) **[20]**

QUESTION 4: BOX GUTTERS AND CHIMNEY FLASHING

4.1 How would you best describe a *box gutter*? (2)

4.2 Name TWO methods that could be used to prevent a reaction between aluminium and freshly placed cement. (2)

4.3 With the aid of a diagram draw a cross section of an abutment flashing. (4)

4.4 Name ONE type of material that can be used to manufacture soakers. (1)

4.5 List SIX steps that should be followed when fabricating a piece for a chimney flashing. (6) **[15]**

QUESTION 5: INSTALL A SOLAR WATER HEATER

5.1 What is a solar collector part of and what purpose does it serve? (2)

5.2 How can radiant energy be distributed? (2)

5.3 Complete the following sentences by filling in the missing word or words. Write only the word or words next to the question number (5.3.1–5.3.8) in the ANSWER BOOK.

Makes us hot, harmful by-products, energy from the sun, run out, life giving energy, feel every day, radiant energy, source of energy.

Solar energy or (5.3.1) … is something we (5.3.2) … The rays of the sun (5.3.3) … because they contain (5.3.4) … It is a primary source of (5.3.5) … and unlike other (5.3.6) …, it does not (5.3.7) … or create (5.3.8)…

(8 × 1) (8)

5.4 Why do solar water heating systems need a back-up system? (2)

5.5 Describe the meaning of the following term: '*standby heat loss.*' (2)

5.6 Which part of an evacuated-tube collector absorbs the solar energy? (1)

5.7 Supply a very important fact that must be considered during the installation of solar heating system. (2)

5.8 List THREE steps that should be followed when installing the solar collector on the roof. (3)

5.9 What is a *demand water heater*? (2)

5.10 Who should check that the roof can hold the weight of a solar hot water cylinder, design any additional strengthening for the roof and sign off on the construction? (1)

[25]

QUESTION 6: BASIC BUSINESS CONCEPTS

6.1 Which important document must be presented to the bank to make a loan for your new business? (1)

6.2 List TWO important questions that an entrepreneur should ask when doing market research about competitors? (2)

6.3 List THREE points of importance that must be specified in a contract drawn up between yourself and a client. (3)

6.4 Give FOUR advantages that a business has, when making use of electronic fund transfers (EFT). (4)

[10]

TOTAL: 100

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